

Calculation of Zero Lift Drag Coefficients Using the Wake Profile Method for NACA 0012 and NACA 4412 Airfoils

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This experiment was used to find the zero lift drag coefficient of NACA 0012 and NACA 4412 airfoils using their measured wake profiles from a low-speed wind tunnel. The hot wire anemometer was calibrated by relating the logarithm of velocity to the logarithm of voltage, producing $a = 10.82$ and $b = -7.01$. Then each airfoil was mounted at the zero-lift angle of attack, and velocity was measured from the wake just behind the trailing edge. By integrating the velocity, the drag came out to be 0.4440 N/m for NACA 0012 and 0.6765 N/m for NACA 4412 with the drag coefficients being $C_D = 0.0064$ and $C_D = 0.0099$. Overall, the computed results matched expectations and displayed how wake profiles can estimate drag of an airfoil.

I. Nomenclature

a	=	exponent constant for King's law
b	=	constant for King's law
c	=	airfoil chord length (m)
$C_{D,0}$	=	zero lift drag coefficient
C_D	=	drag coefficient
D	=	drag force (N)
E	=	hot wire anemometer voltage (V)
q	=	dynamic pressure (Pa)
h_{norm}	=	normalized height ($\frac{h}{h_{\text{max}}}$)
p	=	static pressure (Pa)
q_∞	=	dynamic pressure (Pa)
q_∞	=	freestream dynamic pressure (Pa)
ρ	=	air density (kg/m^3)
U	=	local velocity (m/s)
U_∞	=	freestream velocity (m/s)
U_{norm}	=	normalized velocity ($\frac{U}{U_{\text{max}}}$)
Δh	=	step increment per motor step (m)

II. Introduction

Drag is of major importance in airfoil design and performance. As airfoils are set to zero angle of attack, it still experiences drag which comes from friction from the airfoil and pressure drag from viscous effects and wake formation behind the trailing edge of the airfoil.

In this experiment, the wake profile method is used to find the zero lift drag for the NACA 0012 symmetric airfoil and the NACA 4412 cambered airfoil. This method is based on momentum theorem that relates to the drag on a body to loss of momentum in the wake of the airfoil. As mentioned [1], the drag force is directly linked to the size and depth of the wake, so the greater the velocity deficit in the wake, the greater the drag. So, by measuring the velocity profile with a hot wire anemometer, the drag can be found.

This experiment was done in a low-speed wind tunnel with a pitot tube, pressure transducer, stepper motor, and a hot wire anemometer. The pitot tube measured the static pressure to determine the freestream velocity while the pressure transducer converted the pressure differences to voltage signals. The stepper motor moved the hot wire probe precisely down the airfoil's wake. Finally, the hot wire anemometer was used to measure velocity data from the wake of the airfoil. This is done by passing an electrical current through a wire that heats up and as air flows over the wire, it cools due to convection. Then energy is consumed to heat the wire back up, which can be related to the airflow velocity allowing for voltage to be converted to velocity with calibration.

In summary, the experiment shows how drag can be measured through wake measurement. When comparing the results of NACA 0012 and NACA 4412 airfoils show how an airfoil with camber has increased drag by changing the pressure distribution and wake shape leading to a velocity decrease and larger energy loss [2].

III. Procedure

Start the experiment with no airfoil installed and place the hot wire anemometer probe in the center of the wind tunnel where airflow is uniform. Then turn on the wind tunnel and set the fan frequency from 10 Hz to 50 Hz in 5 Hz increments. At each speed the hot wire voltage E and freestream velocity U measured from the pitot tube were recorded. The data was then plotted $\ln(U)$ vs $\ln(E)$ and performed linear regression to determine the constants a and b used in King's law (Eq. (1)). The calibration slope from King's Law allows for the conversion of wake voltage data to velocity for later measurements.

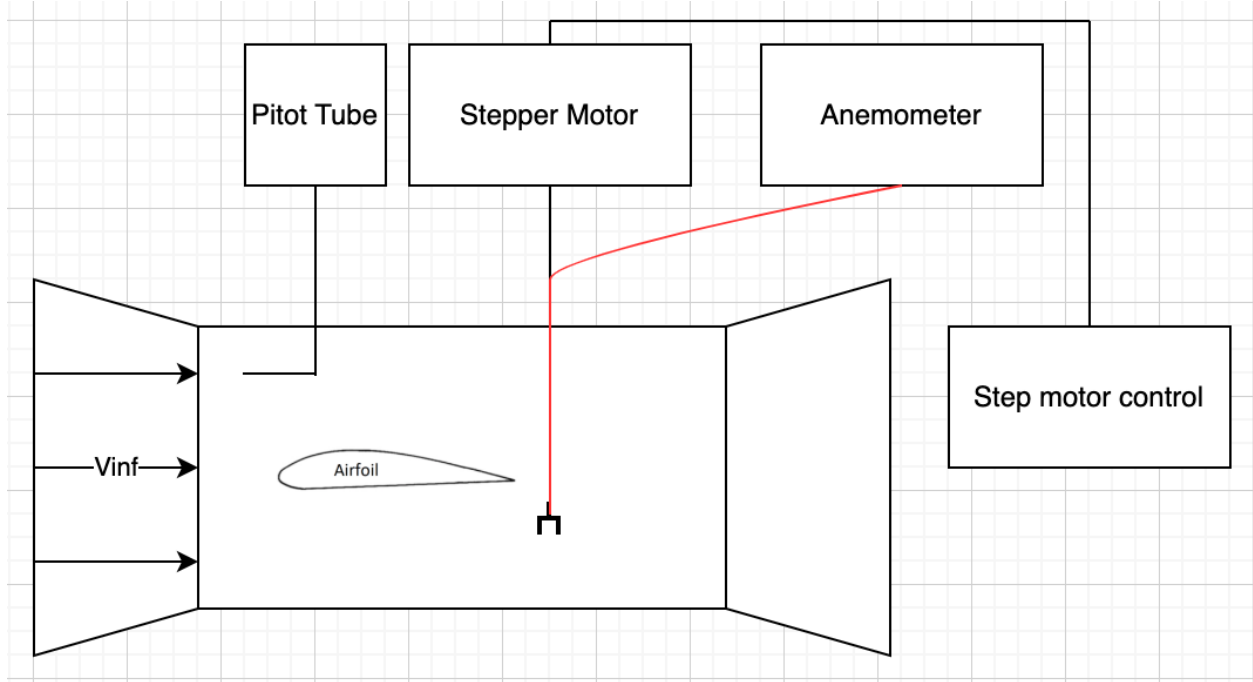


Fig. 1 Diagram of the wind tunnel setup

$$U = e^b E^a \quad (1)$$

After the data calibration, is completed turn off the wind tunnel and install NACA 0012 airfoil at its zero-lift angle of attack. Then place the hot wire anemometer probe $\frac{1}{4}$ inch behind the trailing edge of the airfoil and zero at bottom height ready to move with the stepper motor up increasing steps (never decrease steps or else data collection will become inaccurate) use fig.1 as reference. Then set the wind tunnel speed to 40 Hz and move the probe upward through the airfoil wake with smaller steps (100-200) near the centerline of the airfoil and larger step (400-1000) as the probe moves farther from the centerline. Using stepper motor info related to the specific device (16 revolutions per inch and 400 steps per revolution) the vertical movement per step can be found using (Eq. (2)). Finally repeat this process for the NACA 4412 airfoil at its zero-lift angle of attack of negative four degrees.

$$\Delta h = \frac{0.0254}{\left(\frac{rev}{in}\right)\left(\frac{steps}{rev}\right)} \quad (2)$$

Now for drag calculation a control volume is applied around the airfoil wake as seen in fig. 2. The left side of the control volume represents U_{∞} while the right-side changes due to drag from the airfoil.

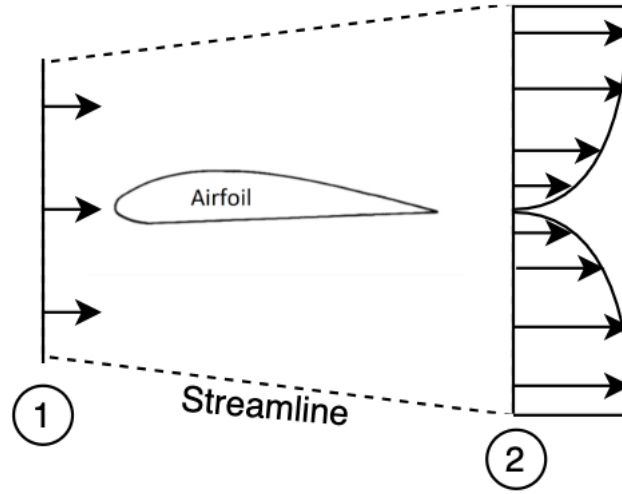


Fig. 2 control volume for drag calculation

Under the assumptions of steady, incompressible, inviscid drag for unit span was found by the absence of momentum across the wake (Eq. (3)).

$$D' = \rho \int_{-\infty}^{\infty} u(U_{\infty} - u)dy \quad (3)$$

Air density was found using the average dynamic pressure and freestream velocity as seen in (Eq. (4))

$$\rho = \frac{2q}{U_{\infty}^2} \quad (4)$$

Then the freestream dynamic pressure was found with (Eq. (5)).

$$q_{\infty} = \frac{1}{2} \rho U_{\infty}^2 \quad (5)$$

Finally (Eq. (6)) was used to find the drag coefficient at zero lift.

$$C_{D,0} = \frac{D'}{q_{\infty} c} \quad (6)$$

IV. Results

The Hot wire anemometer calibration between voltage and air velocity was linearized in log-to-log form and fit with least squares regression to calculate the constants a and b for the King's Law:

$$a = 10.8158, b = -7.0070$$

The calibration curve Fig. 3 shows the tight linear relationship between $\ln(E)$ and $\ln(U)$ giving an accurate method to convert voltage to velocity.

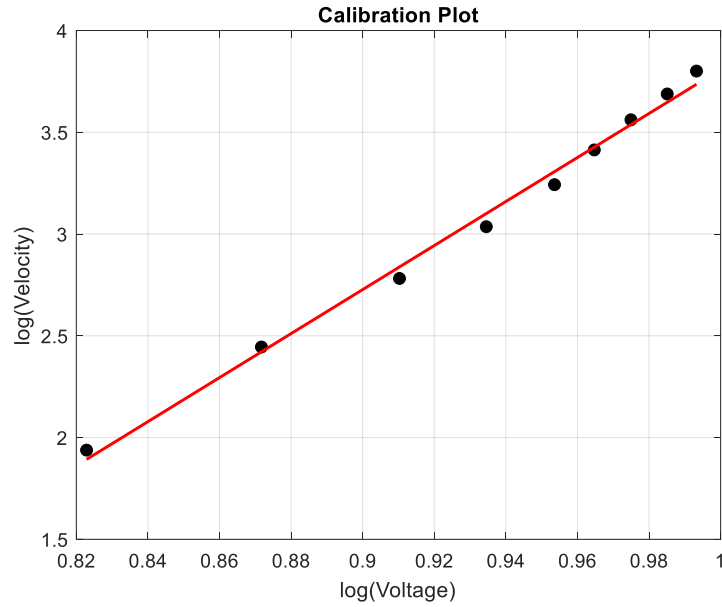


Fig. 3 Diagram of the wind tunnel setup

Fig. 4 and Fig. 5 both display the normalized velocity profiles behind the NACA airfoils. With the x axis being nondimensional velocity $\frac{U}{U_\infty}$ and y axis being normalized as $\frac{h}{h_{max}}$. For the NACA 0012 airfoil Fig. 4 the wake profile has a small thin deficit at the centerline of the airfoil. While the NACA 4412 airfoil Fig. 5 has a deeper and wider velocity deficit due to its cambered shape.

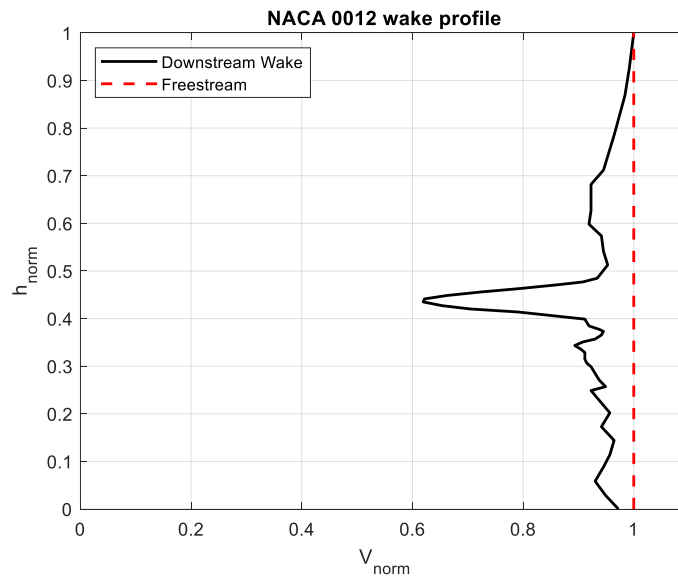


Fig. 4 NACA 0012 Normalized Wake Profile

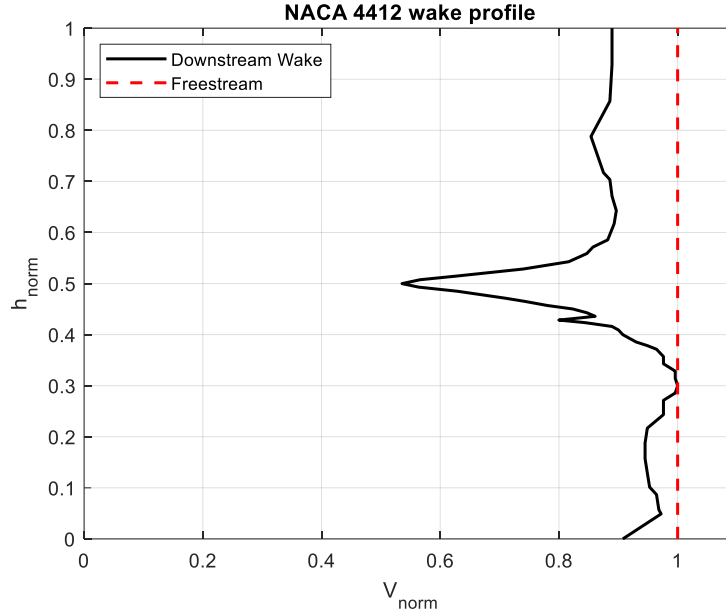


Fig. 5 NACA 4412 Normalized Wake Profile

To Compute the zero lift, drag coefficient and drag for unit span D' , compare momentum entering and leaving the airfoil wake. Air density was then calculated with measured dynamic pressure and the freestream velocity. This resulted in the drag forces and zero lift drag coefficients below:

Airfoil	U_{∞} (m/s)	D' (N/m)	C_{D0}
NACA 0012	41.921	0.4440	0.0064
NACA 4412	40.597	0.6765	0.0090

The NACA 0012 had a lower zero lift drag coefficient matching Fig. 4 with a thinner wake and less viscous loss. The cambered NACA 4412 produced higher drag due to having a larger wake width and spike at the centerline of the airfoil. This experiment aligns with aerodynamic theory that a camber increases pressure drag when at zero lift. These results also match closely with Abbott and von Doenhoff which reads about 0.006 for NACA 0012 and 0.009 to 0.010 for NACA 4412.

V. Conclusion

Using the wake profile method successfully computed the zero lift drag coefficients of the NACA 0012 and NACA 4412 airfoils. With the momentum deficit approach, the drag coefficients came out to be $C_{D,0} = 0.0064$ for the NACA 0012 and $C_{D,0} = 0.0099$ for the NACA 4412. These values accurately matched with the Abbott and von Doenhoff, values proving the validity of the experiment results.

For this experiment the assumptions of steady, incompressible, inviscid flow and uniform helped measure the wake velocity profile behind the airfoil. Doing this simplifies the computations but introduces possibility of small errors in calculations. Other possibilities of uncertainty may have resulted from vibration in the hot wire anemometer probe that increases as the stepper motor is increased. A method to fix this could be increasing the radius of the probe but as this increases it may start to disrupt flow and cause more uncertainty in results.

Overall, the results match all expectations with the NACA 4412 cambered airfoil giving greater drag than the symmetric NACA 0012 airfoil since having camber increases the surface area generating higher skin friction drag. Therefore, this method effectively linked velocity of the wake to the drag produced by the airfoil. The results also confirm that while cambered airfoils generate greater lift, they have a tradeoff of creating higher drag at zero lift. This plays a major role in air-foil design depending on requirements needed for specific aircraft. In conclusion calculating drag from wake measurements is a very effective method that provides accurate results.

VI. Appendix

Appendix A: Tabulated Data

Table 1: Density

Airfoil	U_{∞} (m/s)	q_{∞}	$\rho(\frac{kg}{m^3})$
NACA 0012	41.921	691.506	0.7870
NACA 4412	40.597	685.407	0.8318

Table 2: Calibration Data

E (V)	U (m/s)	ln(E)	ln(U)
2.2770	6.947	0.8229	1.9383
2.3910	11.532	0.8717	2.4451
2.4850	16.148	0.9103	2.7818
2.5460	20.822	0.9345	3.0360
2.5950	25.598	0.9536	3.2425
2.6240	30.367	0.9647	3.4134
2.6510	35.210	0.9749	3.5613
2.6780	39.974	0.9851	3.6882
2.7000	44.737	0.9933	3.8008

Table 3: NACA 0012 Normalized Velocity Data

h/h_max	U/U_max
0.000	0.972
0.028	0.949
0.058	0.930
0.088	0.945
0.114	0.957
0.144	0.965
0.173	0.942
0.202	0.957
0.249	0.923
0.257	0.949
0.271	0.938
0.285	0.930
0.299	0.923
0.306	0.915
0.315	0.912
0.322	0.912
0.329	0.912
0.336	0.904
0.343	0.893
0.351	0.908
0.357	0.930
0.365	0.942
0.373	0.945
0.377	0.938

0.385	0.919
0.399	0.912
0.414	0.791
0.420	0.705
0.427	0.654
0.435	0.619
0.441	0.622
0.448	0.662
0.456	0.725
0.463	0.791
0.471	0.861
0.477	0.908
0.485	0.934
0.513	0.953
0.541	0.945
0.574	0.942
0.598	0.919
0.626	0.923
0.682	0.923
0.712	0.945
0.785	0.965
0.869	0.984
0.926	0.992
1.000	1.000

Table 4: NACA 4412 Normalized Velocity Data

h/h_{max}	U/U_{max}
0.000	0.908
0.049	0.972
0.057	0.968
0.087	0.964
0.101	0.953
0.129	0.949
0.157	0.945
0.188	0.945
0.217	0.949
0.243	0.976
0.271	0.976
0.286	0.996
0.300	1.000
0.314	0.996
0.329	0.996
0.343	0.976
0.357	0.976
0.371	0.964
0.378	0.949

0.386	0.930
0.393	0.919
0.400	0.908
0.409	0.900
0.416	0.889
0.424	0.844
0.428	0.800
0.436	0.861
0.443	0.847
0.450	0.823
0.457	0.780
0.465	0.743
0.471	0.710
0.485	0.629
0.493	0.564
0.500	0.536
0.507	0.566
0.514	0.626
0.528	0.740
0.543	0.816
0.559	0.847
0.571	0.857
0.586	0.882
0.617	0.893
0.643	0.897
0.671	0.889
0.704	0.886
0.717	0.875
0.788	0.854
0.857	0.886
0.928	0.889
1.000	0.889

Table 5: C_D values

Airfoil	U_∞ (m/s)	D'(N/m)	$C_{D,0}$ (calculated)	$C_{D,0}$ (Abbot's)
0012	41.921	0.4440	0.0064	0.0060
4412	40.597	0.6765	0.0099	0.0095

Appendix B: MATLAB Code

CODE:

```
clc; clear; close all;

% AEE 360 Lab 3

% Files
fCal = 'Fri_B_730_calibration.txt';
f0012 = 'Fri_B_730_0012 (1).txt';
f4412 = 'Fri_B_730_4412.txt';

% Load
Tcal = importdata(fCal);
D12 = importdata(f0012);
D44 = importdata(f4412);

% Extract
% Calibration
E_cal = Tcal.data(:,3); % hot-wire voltage [V]
U_cal = Tcal.data(:,4); % velocity [m/s]

% 0012
q12 = D12.data(:,2); % dynamic pressure [Pa]
E12 = D12.data(:,3); % hot-wire voltage [V]
st12 = D12.data(:,6); % motor steps

% 4412
q44 = D44.data(:,2); % dynamic pressure [Pa]
E44 = D44.data(:,3); % hot-wire voltage [V]
st44 = D44.data(:,6); % motor steps

% step motor constants
revPerIn = 16; % rev / inch
stepsRev = 400; % steps / rev
cmPerStep = 2.54 / (revPerIn*stepsRev);
mPerStep = 0.0254 / (revPerIn*stepsRev);

% King's law calc
x = log(E_cal);
y = log(U_cal);

A = [x, ones(size(x))];
p = A \ y;
a = p(1);
b = p(2);

yfit = A*p;
```

```

fprintf(' a = %.6f\n b = %.6f\n', a, b);

figure;
plot(x, y, 'ko','MarkerFaceColor','k');
hold on;
grid on;
box on;
plot(x, yfit, 'r-','LineWidth',1.5);
xlabel('ln(Voltage)');
ylabel('ln(Velocity)');
title('Hot-Wire Calibration (log-log)');
legend('Data','Linear Fit','Location','northwest');

% NACA 0012
U12 = exp(b) .* (E12.^a); % voltage to velocity
y12_cm = st12*cmPerStep; % step to distance
y12_m = st12*mPerStep;

% sort height
[y12_m, idx12] = sort(y12_m);
y12_cm = y12_cm(idx12);
U12 = U12(idx12);
q12 = q12(idx12);

% normalize
y12n = y12_cm / max(y12_cm);
U12n = U12 / max(U12);

% figure
figure;
plot(U12n, y12n, 'k-','LineWidth',1.5);
hold on;
grid on;
box on;
plot([1 1],[0 1],'r--','LineWidth',1.5);
xlabel('V_{norm}');
ylabel('h_{norm}');
title('NACA 0012 wake profile');
xlim([0 1.1]);
ylim([0 1]);
legend('Downstream Wake','Freestream','Location','northwest');

% NACA 4412
U44 = exp(b) .* (E44.^a);
y44_cm = st44*cmPerStep;
y44_m = st44*mPerStep;

[y44_m, idx44] = sort(y44_m);

```

```

y44_cm = y44_cm(idx44);
U44 = U44(idx44);
q44 = q44(idx44);

y44n = y44_cm / max(y44_cm);
U44n = U44 / max(U44);

figure;
plot(U44n, y44n, 'k-', 'LineWidth', 1.5);
hold on;
grid on;
box on;
plot([1 1], [0 1], 'r--', 'LineWidth', 1.5);
xlabel('V_{norm}');
ylabel('h_{norm}');
title('NACA 4412 wake profile');
xlim([0 1.1]);
ylim([0 1]);
legend('Downstream Wake', 'Freestream', 'Location', 'northwest');

```

```

%% Drag calc

```

```

c = 0.10; % chord [m]

```

```

% 0012
Uinf12 = max(U12);
h2_12 = max(y12_m);
mOut12 = trapz(y12_m, U12);
h1_12 = mOut12 / Uinf12;
rho12 = mean(2*q12) / (Uinf12^2);
Pout12 = trapz(y12_m, U12.^2);
Pin12 = Uinf12^2 * h1_12;
D12 = (rho12 * (Pin12 - Pout12)) * c;
qinf12 = 0.5 * rho12 * Uinf12^2;
Cd12 = D12 / (qinf12 * c);

fprintf('\nNACA 0012 drag\n');
fprintf('U_inf = %.3f m/s\n', Uinf12);
fprintf('D = %.4f N\n', D12);
fprintf('C_D0 = %.4f\n', Cd12);

```

```

% 4412
Uinf44 = max(U44);
h2_44 = max(y44_m);
mOut44 = trapz(y44_m, U44);
h1_44 = mOut44 / Uinf44;
rho44 = mean(2*q44) / (Uinf44^2);
Pout44 = trapz(y44_m, U44.^2);

```

```

Pin44 = Uinf44^2 * h1_44;
D44 = (rho44 * (Pin44 - Pout44)) * c;
qinf44 = 0.5 * rho44 * Uinf44^2;
Cd44 = D44 / (qinf44 * c);

fprintf('NACA 4412 drag\n');
fprintf(' U_inf = %.3f m/s\n', Uinf44);
fprintf(' D = %.4f N\n', D44);
fprintf(' C_D0 = %.4f\n', Cd44);

fprintf('\nTable A-1: Density\n');
fprintf('airfoil\tU_inf (m/s)\tq_inf (Pa)\trho (kg/m^3)\n');
qinf12 = 0.5*rho12*Uinf12^2;
qinf44 = 0.5*rho44*Uinf44^2;
fprintf('NACA0012\t%.3f\t%.3f\t%.4f\n', Uinf12, qinf12, rho12);
fprintf('NACA4412\t%.3f\t%.3f\t%.4f\n', Uinf44, qinf44, rho44);

fprintf('\nTable A-2: Calibration Data\n');
fprintf('a = %.6f, b = %.6f\n', a, b);
fprintf('E (V)\tU (m/s)\tln(E)\tln(U)\n');
for k = 1:numel(E_cal)
    fprintf('%.4f\t%.3f\t%.4f\t%.4f\n', E_cal(k), U_cal(k), log(E_cal(k)), log(U_cal(k)));
end

fprintf('\nTable A-3: NACA 0012 Normalized Velocity Data\n');
fprintf('h/h_max\tU/U_max\n');
for k = 1:numel(y12n)
    fprintf('%.3f\t%.3f\n', y12n(k), U12n(k));
end

fprintf('\nTable A-4: NACA 4412 Normalized Velocity Data\n');
fprintf('h/h_max\tU/U_max\n');
for k = 1:numel(y44n)
    fprintf('%.3f\t%.3f\n', y44n(k), U44n(k));
end

% Abbott & von Doenhoff
Cd0_0012_Abb = 0.0060;
Cd0_4412_Abb = 0.0095;

err0012 = 100*(Cd12 - Cd0_0012_Abb)/Cd0_0012_Abb;
err4412 = 100*(Cd44 - Cd0_4412_Abb)/Cd0_4412_Abb;

fprintf('\nTable A-5: Zero-Lift Drag Coefficients\n');
fprintf('Airfoil\tU_inf (m/s)\tD" (N/m)\tC_D0 (calc)\tC_D0 (Abb)\n');

```

```
fprintf('0012\t%.3f\t%.4f\t%.4f\t%.4f\t\n', Uinf12, D12, Cd12, Cd0_0012_Abb);  
fprintf('4412\t%.3f\t%.4f\t%.4f\t%.4f\t\n', Uinf44, D44, Cd44, Cd0_4412_Abb);
```

VII. Acknowledgements

I'd like to thank the laboratory instructors, teaching assistants for their advice and help throughout the lab experiment and report. Also, I would like to thank my groupmates for their assistance with collecting data accurately.

VIII. References

- [1] Brennen, C. E., "The Connection Between Drag and the Wake," *An Internet Book on Fluid Dynamics*, California Institute of Technology, Pasadena, CA, 2015.
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https://www.mouser.com/catalog/additional/ATS_Qpedia_Dec07_Understanding%20hot%20wire%20amemometry9.pdf
- [3] Arizona State University, "AEE 360 Laboratory 3: Drag on a NACA 0012 and 4412 from Wake-Profile Measurements," Course Materials, Canvas, Fall 2025. <https://canvas.asu.edu/courses/235388/modules>